

HEALTHCARE APPOINTMENT NO-SHOW ANALYTICS USING SNOWFLAKE

Business Scenario

A hospital network wants to analyze patient appointment behavior.

The hospital wants to know:

- How many appointments are booked?
- How many patients miss their appointments?
- Which age group has the highest no-show rate?
- Does SMS reminder reduce no-show?
- Which neighborhood has more missed appointments?
- How can Snowflake automatically load new appointment files?
- How can Power BI show the final dashboard?

You are asked to build a Snowflake data pipeline from raw CSV data to a clean reporting table.

Part A: Assignment Questions for Students

Task 1: Create Snowflake Objects

Create:

- Warehouse
- Database
- Schema
- File format
- Internal stage

Use the name:

- HEALTHCARE_WH
- HEALTHCARE_DB
- APPOINTMENT_SCHEMA

ANS:

Creating a warehouse HEALTHCARE_WH

```
1 CREATE WAREHOUSE IF NOT EXISTS HEALTHCARE_WH
2     WAREHOUSE_SIZE = 'XSMALL'
3     AUTO_SUSPEND = 60
4     AUTO_RESUME = TRUE
5     INITIALLY_SUSPENDED = FALSE;
```

Creating a database HEATHCARE_DB

```
7 CREATE DATABASE IF NOT EXISTS HEALTHCARE_DB;
```

Creating a schema APPOINTMENT_SCHEMA

```
11 CREATE SCHEMA IF NOT EXISTS APPOINTMENT_SCHEMA;
```

Creating a file format CSV_FORMAT

```
15 CREATE FILE FORMAT IF NOT EXISTS CSV_FORMAT
16     TYPE = 'CSV'
17     FIELD_OPTIONALLY_ENCLOSED_BY = ''
18     SKIP_HEADER = 1
19     NULL_IF = ('NULL', 'null', '')
20     EMPTY_FIELD_AS_NULL = TRUE
21     TRIM_SPACE = TRUE;
```

Creating a stage APPOINTMENT_STAGE

```
23 CREATE STAGE IF NOT EXISTS APPOINTMENT_STAGE
24     FILE_FORMAT = CSV_FORMAT;
```

It was successfully executed.

Results (9 minutes ago)

TableChartPivot

🔍 📄

1 row ⓘ

74ms

↶ ↷ ⌛

id	status
1	Warehouse HEALTHCARE_WH successfully created.

Results (just now)

TableChartPivot

🔍 📄

1 row ⓘ

58ms

↶ ↷ ⌛

id	name	state	type	size	# running	# queued	is_default	is_current
1	HEALTHCARE_WH	SUSPENDED	STANDARD	X-Small	0	0	N	Y

Results (just now)

TableChartPivot

🔍 📄

1 row ⓘ

78ms

↶ ↷ ⌛

id	status
1	Database HEALTHCARE_DB successfully created.

Results (just now)

TableChartPivot

🔍 📄

1 row ⓘ

46ms

↶ ↷ ⌛

id	created_on	name	is_default	is_current	origin	owner	comment	options
1	2026-05-22 22:35:49	HEALTHCARE_DB	N	Y		ACCOUNTADMIN		

Results (just now)

TableChartPivot

🔍 📄

1 row ⓘ

41ms

↶ ↷ ⌛

id	status
1	Schema APPOINTMENT_SCHEMA successfully created.

Results (just now)								
Table Chart Pivot 1 row 34ms								
	created_on	name	is_default	is_current	database_name	owner	comment	options
1	2026-05-22 22:36:00	APPOINTMENT_SCHEM	N	Y	HEALTHCARE_DB	ACCOUNTADMIN		

Results (just now)	
Table Chart Pivot 1 row 38ms	
	status
1	File format CSV_FORMAT successfully created.

Results (just now)								
Table Chart Pivot 1 row 31ms								
	created_on	name	database_name	schema_name	type	owner	comment	format_opti
1	2026-05-22 22:36:13	CSV_FORMAT	HEALTHCARE_DB	APPOINTMENT_SCHEM	CSV	ACCOUNTADMIN		("TYPE":"CSV"

Results (just now)	
Table Chart Pivot 1 row 88ms	
	status
1	Stage area APPOINTMENT_STAGE successfully created.

Results (just now)								
Table Chart Pivot 1 row 41ms								
	created_on	name	database_name	schema_name	url	has_credentials	has_encryption_key	owner
1	2026-05-22 22:36:24	APPOINTMENT_STAGE	HEALTHCARE_DB	APPOINTMENT_SCHEM		N	N	ACCOUNTA

Task 2: Scale Warehouse Up and Down

Perform the following:

Create warehouse as XSMALL

Scale it up to SMALL

Scale it down again to XSMALL

Suspend the warehouse manually

Resume the warehouse manually

ANS:

Scaling warehouse HEATHCARE_WH to SMALL

```
1  ALTER WAREHOUSE HEALTHCARE_WH SET WAREHOUSE_SIZE = 'SMALL';
```

Scaling warehouse HEATHCARE_WH to XSMALL

```
2  ALTER WAREHOUSE HEALTHCARE_WH SET WAREHOUSE_SIZE = 'XSMALL';
```

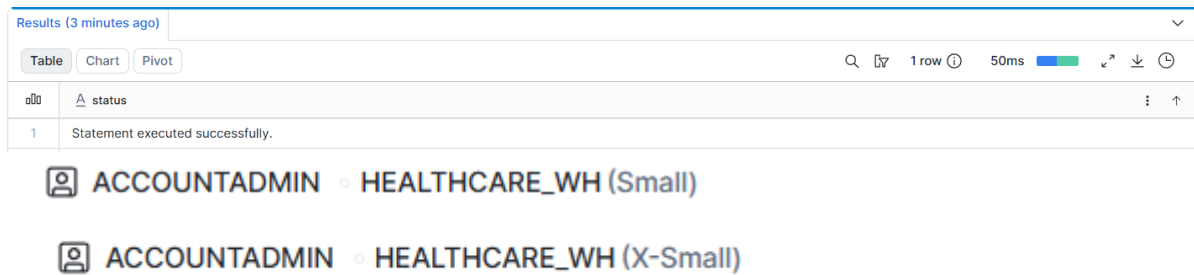
Suspending warehouse HEATHCARE_WH

```
3  ALTER WAREHOUSE HEALTHCARE_WH SUSPEND;
```

Resuming warehouse HEATHCARE_WH

```
4  ALTER WAREHOUSE HEALTHCARE_WH RESUME;
```

It was successfully executed.



The screenshot shows a Snowflake query results window. At the top, it says 'Results (3 minutes ago)'. Below that are tabs for 'Table', 'Chart', and 'Pivot'. To the right of these tabs are icons for search, filters, and a summary: '1 row' and '50ms'. A progress bar is also visible. The main table has one column named 'status' and one row with the value 'Statement executed successfully.'. Below the table, there are two database and schema paths: 'ACCOUNTADMIN - HEALTHCARE_WH (Small)' and 'ACCOUNTADMIN - HEALTHCARE_WH (X-Small)'.

	status
1	Statement executed successfully.

Task 3: Download Dataset from Open URL

Use this public CSV dataset:

<https://raw.githubusercontent.com/kb22/Medical-Appointment-No-Show-Prediction/master/data/dataset.csv>

Download the CSV file to your local system.

File name:

appointment_no_show.csv

ANS:

Downloaded the CSV file as appointment_no_show.csv in the local system

Task 4: Upload CSV into Snowflake Stage

Upload the CSV file into Snowflake internal stage using Snowsight.

Stage name:

APPOINTMENT_STAGE

ANS:

Uploading into Snowflake internal stage APPOINTMENT_STAGE using Snowsight.

Upload Your Files

1 file → APPOINTMENT_STAGE

appointment_no_show.csv - 10.2MB

Select database, schema and stage

Schema HEALTHCARE_DB.APPOINTMENT_SCHEMA APPOINTMENT_STAGE

Specify the path to an existing folder or create a new one (optional)

Example / Path / To / Folder

Once the files finish uploading, your Directory Table will automatically refresh.

Close

Upload

It was successfully executed.

Results (just now)				
Table Chart Pivot				
1 row 41ms				
#	name	size	md5	last_modified
1	appointment_stage/appointment_no_show.csv	10739536	30a3110cac08131e4b69c382a3783b9	Sat, 23 May 2026 06:11:09 GMT

Results (just now)		
Table Chart Pivot		
5 rows 1.4s		
#	METADATA\$FILENAME	METADATA\$FILE_ROW_NUMBER
1	appointment_no_show.csv	1
2	appointment_no_show.csv	2
3	appointment_no_show.csv	3
4	appointment_no_show.csv	4
5	appointment_no_show.csv	5

Task 5: Create Raw Table

Create a raw table to store CSV data exactly as received.

Table name:

APPOINTMENT_RAW

ANS:

Creating a raw table APPOINTMENT_RAW

```

5      CREATE OR REPLACE TABLE APPOINTMENT_RAW (
6          PATIENTID          VARCHAR(50) ,
7          APPOINTMENTID      VARCHAR(50) ,
8          GENDER              VARCHAR(10) ,
9          SCHEDULEDDAY        VARCHAR(50) ,
10         APPOINTMENTDAY      VARCHAR(50) ,
11         AGE                  VARCHAR(10) ,
12         NEIGHBOURHOOD        VARCHAR(100) ,
13         SCHOLARSHIP          VARCHAR(5) ,
14         HIPERTENSION          VARCHAR(5) ,
15         DIABETES              VARCHAR(5) ,
16         ALCOHOLISM            VARCHAR(5) ,
17         HANDCAP               VARCHAR(5) ,
18         SMS_RECEIVED          VARCHAR(5) ,
19         NO_SHOW               VARCHAR(5)
20     );

```

It was successfully executed.

Results (just now)

Table Chart Pivot 1 row 119ms

1	Table APPOINTMENT_RAW successfully created.								
---	---	--	--	--	--	--	--	--	--

Results (just now)

Table Chart Pivot 14 rows 40ms

	name	type	kind	null?	default	primary key	unique key	check	expression
1	PATIENTID	VARCHAR(50)	COLUMN	Y	null	N	N	null	null
2	APPOINTMENTID	VARCHAR(50)	COLUMN	Y	null	N	N	null	null
3	GENDER	VARCHAR(10)	COLUMN	Y	null	N	N	null	null
4	SCHEDULEDDAY	VARCHAR(50)	COLUMN	Y	null	N	N	null	null
5	APPOINTMENTDAY	VARCHAR(50)	COLUMN	Y	null	N	N	null	null
6	AGE	VARCHAR(10)	COLUMN	Y	null	N	N	null	null
7	NEIGHBOURHOOD	VARCHAR(100)	COLUMN	Y	null	N	N	null	null
8	SCHOLARSHIP	VARCHAR(5)	COLUMN	Y	null	N	N	null	null
9	HIPERTENSION	VARCHAR(5)	COLUMN	Y	null	N	N	null	null
10	DIABETES	VARCHAR(5)	COLUMN	Y	null	N	N	null	null
11	ALCOHOLISM	VARCHAR(5)	COLUMN	Y	null	N	N	null	null
12	HANDCAP	VARCHAR(5)	COLUMN	Y	null	N	N	null	null
13	SMS_RECEIVED	VARCHAR(5)	COLUMN	Y	null	N	N	null	null
14	NO_SHOW	VARCHAR(5)	COLUMN	Y	null	N	N	null	null

Task 6: Load Data from Stage to Raw Table

Use COPY INTO to load staged CSV data into the raw table. Snowflake COPY INTO <table> loads staged files into a table using the selected file format.

ANS:

Copying CSV file's data into raw table APPOINTMENT_RAW

```

5 COPY INTO APPOINTMENT_RAW
6 FROM @APPOINTMENT_STAGE/appointment_no_show.csv
7 FILE_FORMAT = (FORMAT_NAME = CSV_FORMAT)
8 ON_ERROR = 'CONTINUE';

```

Verifying

```

10 SELECT COUNT(*) AS TOTAL_ROWS_LOADED FROM APPOINTMENT_RAW;
11
12 SELECT * FROM APPOINTMENT_RAW LIMIT 10;

```

It was successfully executed.

Results (just now)

TableChartPivot

🔍🔗1 row ⓘ3.4s

id	file	status	# rows_parsed	# rows_loaded	# error_limit	# errors_seen	first_error	# first_error_id
1	appointment_stage/apj	LOADED	110527	110527	110527	0	null	

Results (just now)

TableChartPivot

🔍🔗1 row ⓘ64ms

id	# TOTAL_ROWS_LOADED
1	110527

Results (just now)

TableChartPivot

🔍🔗10 rows ⓘ67ms

id	PATIENTID	APPOINTMENTID	GENDER	SCHEDULEDDAY	APPOINTMENTDAY	AGE	NEIGHBOURHOOD	SCHOLARS
1	29872499824296	5642903	F	2016-04-29T18:38:08Z	2016-04-29T00:00:00Z	62	JARDIM DA PENHA	0
2	558997776694438	5642503	M	2016-04-29T16:08:27Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0
3	4262962299951	5642549	F	2016-04-29T16:19:04Z	2016-04-29T00:00:00Z	62	MATA DA PRAIA	0
4	867951213174	5642828	F	2016-04-29T17:29:31Z	2016-04-29T00:00:00Z	8	PONTAL DE CAMBURI	0
5	8841186448183	5642494	F	2016-04-29T16:07:23Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0
6	95985133231274	5626772	F	2016-04-27T08:36:51Z	2016-04-29T00:00:00Z	76	REPÚBLICA	0
7	733688164476661	5630279	F	2016-04-27T15:05:12Z	2016-04-29T00:00:00Z	23	GOIABEIRAS	0
8	3449833394123	5630575	F	2016-04-27T15:39:58Z	2016-04-29T00:00:00Z	39	GOIABEIRAS	0
9	56394729949972	5638447	F	2016-04-29T08:02:16Z	2016-04-29T00:00:00Z	21	ANDORINHAS	0
10	78124564369297	5629123	F	2016-04-27T12:48:25Z	2016-04-29T00:00:00Z	19	CONQUISTA	0

Task 7: Create Final Table

Create a transformed final table:

APPOINTMENT_FINAL

The final table should include:

- Patient ID
- Appointment ID
- Gender
- Age
- Age group
- Scheduled date
- Appointment date
- Waiting days

- **Neighborhood**
- **Scholarship**
- **Hypertension**
- **Diabetes**
- **Alcoholism**
- **Handicap**
- **SMS received**
- **No-show status**
- **Attendance status**

ANS:

Creating a final table APPOINTMENT_FINAL

```

5      CREATE OR REPLACE TABLE APPOINTMENT_FINAL (
6          PATIENT_ID          VARCHAR(50) ,
7          APPOINTMENT_ID      VARCHAR(50) ,
8          GENDER              VARCHAR(10) ,
9          AGE                  INT ,
10         AGE_GROUP            VARCHAR(20) ,
11         SCHEDULED_DATE       DATE ,
12         APPOINTMENT_DATE     DATE ,
13         WAITING_DAYS         INT ,
14         NEIGHBOURHOOD        VARCHAR(100) ,
15         SCHOLARSHIP          INT ,
16         HYPERTENSION          INT ,
17         DIABETES              INT ,
18         ALCOHOLISM           INT ,
19         HANDICAP              INT ,
20         SMS_RECEIVED          INT ,
21         NO_SHOW_STATUS       VARCHAR(5) ,
22         ATTENDANCE_STATUS    VARCHAR(20)
23     );

```

It was successfully executed.

Results (just now)	
Table	Chart Pivot
1 row	113ms
1	Table APPOINTMENT_FINAL successfully created.

Results (just now)

Table Chart Pivot 17 rows 39ms

id	name	type	kind	null?	default	primary key	unique key	check	expression
1	PATIENT_ID	VARCHAR(50)	COLUMN	Y	null	N	N	null	null
2	APPOINTMENT_ID	VARCHAR(50)	COLUMN	Y	null	N	N	null	null
3	GENDER	VARCHAR(10)	COLUMN	Y	null	N	N	null	null
4	AGE	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
5	AGE_GROUP	VARCHAR(20)	COLUMN	Y	null	N	N	null	null
6	SCHEDULED_DATE	DATE	COLUMN	Y	null	N	N	null	null
7	APPOINTMENT_DATE	DATE	COLUMN	Y	null	N	N	null	null
8	WAITING_DAYS	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
9	NEIGHBOURHOOD	VARCHAR(100)	COLUMN	Y	null	N	N	null	null
10	SCHOLARSHIP	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
11	HYPERTENSION	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
12	DIABETES	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
13	ALCOHOLISM	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
14	HANDICAP	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
15	SMS_RECEIVED	NUMBER(38,0)	COLUMN	Y	null	N	N	null	null
16	NO_SHOW_STATUS	VARCHAR(5)	COLUMN	Y	null	N	N	null	null
17	ATTENDANCE_STATUS	VARCHAR(20)	COLUMN	Y	null	N	N	null	null

Open or move Cortex Code

Task 8: Use Snowpark Python

Create a Snowpark Python stored procedure to transform data from raw table to final table.

Snowpark Python procedures can run Python logic inside Snowflake and perform queries, updates, and transformations on Snowflake tables.

ANS:

Creating a Snowpark Python procedure TRANSFORM_APPOINTMENTS()

```

5 CREATE OR REPLACE PROCEDURE TRANSFORM_APPOINTMENTS()
6 RETURNS STRING
7 LANGUAGE PYTHON
8 RUNTIME_VERSION = '3.11'
9 PACKAGES = ('snowflake-snowpark-python')
10 HANDLER = 'run'
11 AS
12 $$
13 def run(session):
14     session.sql("TRUNCATE TABLE APPOINTMENT_FINAL").collect()
15
16     raw_df = session.table("APPOINTMENT_RAW")
17
18     from snowflake.snowpark.functions import col, to_date, datediff, when, lit, try_cast
19     from snowflake.snowpark.types import IntegerType
20
21     transformed_df = raw_df.select(
22         col("PATIENTID").alias("PATIENT_ID"),
23         col("APPOINTMENTID").alias("APPOINTMENT_ID"),
24         col("GENDER"),
25         try_cast(col("AGE"), IntegerType()).alias("AGE"),
26         when(try_cast(col("AGE"), IntegerType()) < 13, lit("Child"))
27         .when(try_cast(col("AGE"), IntegerType()) < 18, lit("Teen"))
28         .when(try_cast(col("AGE"), IntegerType()) < 30, lit("Young Adult"))
29         .when(try_cast(col("AGE"), IntegerType()) < 50, lit("Adult"))
30         .when(try_cast(col("AGE"), IntegerType()) < 65, lit("Middle Age"))
31         .otherwise(lit("Senior")).alias("AGE_GROUP"),
32         to_date(col("SCHEDULEDDAY")).alias("SCHEDULED_DATE"),
33         to_date(col("APPOINTMENTDAY")).alias("APPOINTMENT_DATE"),
34         datediff("day", to_date(col("SCHEDULEDDAY")), to_date(col("APPOINTMENTDAY"))).alias("WAITING_DAYS"),
35         col("NEIGHBOURHOOD"),
36         try_cast(col("SCHOLARSHIP"), IntegerType()).alias("SCHOLARSHIP"),
37         try_cast(col("HIPERTENSION"), IntegerType()).alias("HYPERTENSION"),

```

```

38         try_cast(col("DIABETES"), IntegerType()).alias("DIABETES"),
39         try_cast(col("ALCOHOLISM"), IntegerType()).alias("ALCOHOLISM"),
40         try_cast(col("HANDCAP"), IntegerType()).alias("HANDICAP"),
41         try_cast(col("SMS_RECEIVED"), IntegerType()).alias("SMS_RECEIVED"),
42         col("NO_SHOW").alias("NO_SHOW_STATUS"),
43         when(col("NO_SHOW") == lit("No"), lit("Attended"))
44         .otherwise(lit("No-Show")).alias("ATTENDANCE_STATUS")
45     )
46
47     transformed_df.write.mode("append").save_as_table("APPOINTMENT_FINAL")
48
49     count = session.table("APPOINTMENT_FINAL").count()
50     return f"Transform complete. {count} rows loaded into APPOINTMENT_FINAL."
51 
```

Calling the procedure

```
CALL TRANSFORM_APPOINTMENTS();
```

Verifying

```
SELECT COUNT(*) FROM APPOINTMENT_FINAL;  
SELECT * FROM APPOINTMENT_FINAL LIMIT 10;
```

It was successfully executed.

Results (just now)

Table

Chart

Pivot

🔍

📄

1 row

7.1s

↶

↷

🔄

📄	🔗 status
1	Function TRANSFORM_APPOINTMENTS successfully created.

Results (just now)

Table

Chart

Pivot

🔍

📄

1 row

5.3s

↶

↷

🔄

📄	🔗 TRANSFORM_APPOINTMENTS
1	Transform complete. 110527 rows loaded into APPOINTMENT_FINAL.

Results (just now)

Table

Chart

Pivot

🔍

📄

1 row

26ms

↶

↷

🔄

📄	🔗 # COUNT(*)
1	110527

Results (just now)

Table

Chart

Pivot

🔍

📄

10 rows

79ms

↶

↷

🔄

📄	🔗 PATIENT_ID	🔗 APPOINTMENT_ID	🔗 GENDER	🔗 AGE	🔗 AGE_GROUP	🔗 SCHEDULED_DATE	🔗 APPOINTMENT_DATE	🔗 WAITING_DAYS
1	29872499824296	5642903	F	62	Middle Age	2016-04-29	2016-04-29	0
2	558997776694438	5642503	M	56	Middle Age	2016-04-29	2016-04-29	0
3	4262962299951	5642549	F	62	Middle Age	2016-04-29	2016-04-29	0
4	867951213174	5642828	F	8	Child	2016-04-29	2016-04-29	0
5	8841186448183	5642494	F	56	Middle Age	2016-04-29	2016-04-29	0
6	95985133231274	5626772	F	76	Senior	2016-04-27	2016-04-29	2
7	733688164476661	5630279	F	23	Young Adult	2016-04-27	2016-04-29	2
8	3449833394123	5630575	F	39	Adult	2016-04-27	2016-04-29	2
9	56394729949972	5638447	F	21	Young Adult	2016-04-29	2016-04-29	0
10	78124564369297	5629123	F	19	Young Adult	2016-04-27	2016-04-29	2

Task 9: Create Stream

Create a stream on the raw table to track newly inserted rows.

A Snowflake stream tracks row-level changes in a source table between transactional points, which is useful for change data capture.

ANS:

Creating a stream APPOINTMENT_RAW_STREAM

```
5 CREATE OR REPLACE STREAM APPOINTMENT_RAW_STREAM
6 ON TABLE APPOINTMENT_RAW
7 APPEND_ONLY = TRUE
8 COMMENT = 'Stream to track new rows inserted into APPOINTMENT_RAW';
```

Verifying

```
10 SHOW STREAMS LIKE 'APPOINTMENT_RAW_STREAM';
11
12 SELECT SYSTEM$STREAM_HAS_DATA('APPOINTMENT_RAW_STREAM') AS HAS_NEW_DATA;
```

It was successfully executed.

Results (just now)

Table

Chart

Pivot

🔍

📄

1 row ⓘ

220ms

↶ ↷

🕒

📄

🔗 status

1

Stream APPOINTMENT_RAW_STREAM successfully created.

Results (just now)

Table

Chart

Pivot

🔍

📄

1 row ⓘ

57ms

↶ ↷

🕒

📄

🕒 created_on

🔗 name

🔗 database_name

🔗 schema_name

🔗 owner

🔗 comment

🔗 table_name

🔗 source_type

1

2026-05-23 00:11:15

APPOINTMENT_RAW_S

HEALTHCARE_DB

APPOINTMENT_SCHEM

ACCOUNTADMIN

Stream to track new ro

HEALTHCARE_DB.APP

Table

Results (just now)

Table

Chart

Pivot

🔍

📄

1 row ⓘ

66ms

↶ ↷

🕒

📄

0|1 HAS_NEW_DATA

1

FALSE

Task 10: Use Stream for Incremental Loading

Insert new records into the raw table.

Then use the stream to load only new records into the final table.

ANS:

Inserting into raw table APPOINTMENT_RAW new data

```
5 INSERT INTO APPOINTMENT_RAW (
6 PATIENTID, APPOINTMENTID, GENDER, SCHEDULEDDAY, APPOINTMENTDAY,
7 AGE, NEIGHBOURHOOD, SCHOLARSHIP, HIPERTENSION, DIABETES,
8 ALCOHOLISM, HANDCAP, SMS_RECEIVED, NO_SHOW
9 ) VALUES
10 ('9999999901', 'APT100001', 'F', '2016-05-01T08:00:00Z', '2016-05-15T00:00:00Z', '34', 'JARDIM DA PENHA', '0', '0', '0', '0', '1',
11 'No'),
12 ('9999999902', 'APT100002', 'M', '2016-05-02T09:30:00Z', '2016-05-20T00:00:00Z', '52', 'MARIA ORTIZ', '0', '1', '0', '0', '0', 'Yes'),
13 ('9999999903', 'APT100003', 'F', '2016-05-03T11:00:00Z', '2016-05-18T00:00:00Z', '7', 'RESISTÊNCIA', '1', '0', '0', '0', '0', '1', 'No');
```

Inserting into final table APPOINTMENT_FINAL loaded data in the stream

```

18 INSERT INTO APPOINTMENT_FINAL (
19     PATIENT_ID, APPOINTMENT_ID, GENDER, AGE, AGE_GROUP,
20     SCHEDULED_DATE, APPOINTMENT_DATE, WAITING_DAYS,
21     NEIGHBOURHOOD, SCHOLARSHIP, HYPERTENSION, DIABETES,
22     ALCOHOLISM, HANDICAP, SMS_RECEIVED, NO_SHOW_STATUS, ATTENDANCE_STATUS
23 )
24 SELECT
25     s.PATIENTID,
26     s.APPOINTMENTID,
27     s.GENDER,
28     TRY_CAST(s.AGE AS INT),
29     CASE
30         WHEN TRY_CAST(s.AGE AS INT) < 13 THEN 'Child'
31         WHEN TRY_CAST(s.AGE AS INT) < 18 THEN 'Teen'
32         WHEN TRY_CAST(s.AGE AS INT) < 30 THEN 'Young Adult'
33         WHEN TRY_CAST(s.AGE AS INT) < 50 THEN 'Adult'
34         WHEN TRY_CAST(s.AGE AS INT) < 65 THEN 'Middle Age'
35         ELSE 'Senior'
36     END,
37     TO_DATE(s.SCHEDULEDDAY),
38     TO_DATE(s.APPOINTMENTDAY),
39     DATEDIFF('day', TO_DATE(s.SCHEDULEDDAY), TO_DATE(s.APPOINTMENTDAY)),
40     s.NEIGHBOURHOOD,
41     TRY_CAST(s.SCHOLARSHIP AS INT),
42     TRY_CAST(s.HIPERTENSION AS INT),
43     TRY_CAST(s.DIABETES AS INT),
44     TRY_CAST(s.ALCOHOLISM AS INT),
45     TRY_CAST(s.HANDICAP AS INT),
46     TRY_CAST(s.SMS_RECEIVED AS INT),
47     s.NO_SHOW,
48     CASE WHEN s.NO_SHOW = 'No' THEN 'Attended' ELSE 'No-Show' END
49 FROM APPOINTMENT_RAW_STREAM s
50 WHERE s.METADATA$ACTION = 'INSERT';

```

It was successfully executed.

Results (just now)	
Table	Chart Pivot
1 row	1.1s
# number of rows inserted	
1	3

Results (just now)	
Table	Chart Pivot
1 row	81ms
0 1 HAS_NEW_DATA	
1	TRUE

Results (just now)								
Table	Chart	Pivot	3 rows	142ms				
PATIENTID	APPOINTMENTID	GENDER	SCHEDULEDDAY	APPOINTMENTDAY	AGE	NEIGHBOURHOOD	SCHOLARSHIP	
9999999901	APT100001	F	2016-05-01T08:00:00Z	2016-05-15T00:00:00Z	34	JARDIM DA PENHA	0	
9999999902	APT100002	M	2016-05-02T09:30:00Z	2016-05-20T00:00:00Z	52	MARIA ORTIZ	0	
9999999903	APT100003	F	2016-05-03T11:00:00Z	2016-05-18T00:00:00Z	7	RESISTÊNCIA	1	

Results (just now)	
Table	Chart Pivot
1 row	325ms
# number of rows inserted	
1	3

Results (just now)

Table Chart Pivot

1 row 69ms

0	1 HAS_NEW_DATA
1	FALSE

Results (just now)

Table Chart Pivot

1 row 50ms

0	# TOTAL_FINAL
1	110530

Results (just now)

Table Chart Pivot

5 rows 226ms

PATIENT_ID	APPOINTMENT_ID	GENDER	AGE	AGE_GROUP	SCHEDULED_DATE	APPOINTMENT_DATE	WAITING_DAYS
9999999903	APT100003	F	7	Child	2016-05-03	2016-05-18	15
9999999902	APT100002	M	52	Middle Age	2016-05-02	2016-05-20	18
9999999901	APT100001	F	34	Adult	2016-05-01	2016-05-15	14
99999967293456	5736309	M	59	Middle Age	2016-05-25	2016-06-03	9
9999951146446	5604848	F	38	Adult	2016-04-20	2016-05-04	14

Task 11: Create Snowpipe

Create a Snowpipe pipe to load new CSV files from the stage into the raw table.

Snowpipe loads files as soon as they are available in a stage, using a pipe object that contains a COPY INTO statement.

ANS:

Creating a pipe APPOINTMENT_PIPE

```

5      CREATE OR REPLACE PIPE APPOINTMENT_PIPE
6      AUTO_INGEST = FALSE
7      COMMENT = 'Snowpipe to auto-load CSV files from APPOINTMENT_STAGE into APPOINTMENT_RAW'
8      AS
9      COPY INTO APPOINTMENT_RAW (
10         PATIENTID, APPOINTMENTID, GENDER, SCHEDULEDDAY, APPOINTMENTDAY,
11         AGE, NEIGHBOURHOOD, SCHOLARSHIP, HIPERTENSION, DIABETES,
12         ALCOHOLISM, HANDCAP, SMS_RECEIVED, NO_SHOW
13     )
14     FROM (
15         SELECT
16             $1, $2, $3, $4, $5,
17             $6, $7, $8, $9, $10,
18             $11, $12, $13, $14
19         FROM @APPOINTMENT_STAGE
20     )
21     FILE_FORMAT = (FORMAT_NAME = CSV_FORMAT);

```

It was successfully executed.

Results (just now)

Table Chart Pivot

1 row 170ms

0	status
1	Pipe APPOINTMENT_PIPE successfully created.

Results (just now)

TableChartPivot

🔍🔄1 row ⓘ41ms

↶↷⌵⌴⌚

🗂	📅 created_on	🔗 name	🔗 database_name	🔗 schema_name	🔗 definition	🔗 owner	🔗 notification_channel	🔗 comment
1	2026-05-23 00:25:48	APPOINTMENT_PIPE	HEALTHCARE_DB	APPOINTMENT_SCHEM	COPY INTO APPOINTM	ACCOUNTADMIN	null	Snowpipe t

Results (just now)

TableChartPivot

🔍🔄1 row ⓘ47ms

↶↷⌵⌴⌚

🗂	🔗 SYSTEM\$PIPE_STATUS('APPOINTMENT_PIPE')
1	{"executionState":"RUNNING","pendingFileCount":0,"pendingHistoryRefreshJobsCount":0}

Results (just now)

TableChartPivot

🔍🔄0 rows ⓘ227ms

↶↷⌵⌴⌚

🗂	File	Status
---	------	--------

⚠

Query produced no results

✦

Task 12: Create Reporting View

Create a view for Power BI:

VW_APPOINTMENT_DASHBOARD

ANS:

Creating a view VW_APPOINTMENT_DASHBOARD

```

5      CREATE OR REPLACE VIEW VW_APPOINTMENT_DASHBOARD AS
6      SELECT
7          PATIENT_ID,
8          APPOINTMENT_ID,
9          GENDER,
10         AGE,
11         AGE_GROUP,
12         SCHEDULED_DATE,
13         APPOINTMENT_DATE,
14         WAITING_DAYS,
15         NEIGHBOURHOOD,
16         SCHOLARSHIP,
17         HYPERTENSION,
18         DIABETES,
19         ALCOHOLISM,
20         HANDICAP,
21         SMS_RECEIVED,
22         NO_SHOW_STATUS,
23         ATTENDANCE_STATUS,
24         CASE WHEN NO_SHOW_STATUS = 'Yes' THEN 1 ELSE 0 END AS IS_NO_SHOW,
25         CASE WHEN NO_SHOW_STATUS = 'No' THEN 1 ELSE 0 END AS IS_ATTENDED,
26         CASE WHEN SMS_RECEIVED = 1 THEN 'SMS Sent' ELSE 'No SMS' END AS SMS_LABEL,
27         YEAR(APPOINTMENT_DATE) AS APPOINTMENT_YEAR,
28         MONTH(APPOINTMENT_DATE) AS APPOINTMENT_MONTH,
29         MONTHNAME(APPOINTMENT_DATE) AS APPOINTMENT_MONTH_NAME,
30         DAYNAME(APPOINTMENT_DATE) AS APPOINTMENT_DAY_NAME
31     FROM APPOINTMENT_FINAL
32     WHERE AGE >= 0
33           AND APPOINTMENT_DATE IS NOT NULL
34           AND SCHEDULED_DATE IS NOT NULL;

```

Shows for each age group (Child, Teen, Young Adult, Adult, Middle Age, Senior) how many appointments were booked, how many were missed, and the **no-show percentage**.

```

38  SELECT
39      AGE_GROUP,
40      COUNT(*)                               AS TOTAL_APPOINTMENTS,
41      SUM(IS_NO_SHOW)                       AS NO_SHOW_COUNT,
42      ROUND(SUM(IS_NO_SHOW) * 100.0 / COUNT(*), 2) AS NO_SHOW_RATE_PCT
43  FROM VW_APPOINTMENT_DASHBOARD
44  GROUP BY AGE_GROUP
45  ORDER BY NO_SHOW_RATE_PCT DESC;

```

Compares two groups — "SMS Sent" vs "No SMS" — showing the no-show rate for each.

```

47  SELECT
48      SMS_LABEL,
49      COUNT(*)                               AS TOTAL_APPOINTMENTS,
50      SUM(IS_NO_SHOW)                       AS NO_SHOW_COUNT,
51      ROUND(SUM(IS_NO_SHOW) * 100.0 / COUNT(*), 2) AS NO_SHOW_RATE_PCT
52  FROM VW_APPOINTMENT_DASHBOARD
53  GROUP BY SMS_LABEL;

```

Tells the hospital which areas have the worst attendance problem so they can target interventions.

```

55  SELECT
56      NEIGHBOURHOOD,
57      COUNT(*)                               AS TOTAL_APPOINTMENTS,
58      SUM(IS_NO_SHOW)                       AS NO_SHOW_COUNT,
59      ROUND(SUM(IS_NO_SHOW) * 100.0 / COUNT(*), 2) AS NO_SHOW_RATE_PCT
60  FROM VW_APPOINTMENT_DASHBOARD
61  GROUP BY NEIGHBOURHOOD
62  ORDER BY NO_SHOW_COUNT DESC;

```

It was successfully executed.

Results (just now)					
TableChartPivot 1 row162ms					
View VW_APPOINTMENT_DASHBOARD successfully created.					

Results (just now)					
TableChartPivot 1 row94ms					
# TOTAL_RECORDS					
1	110529				

Results (just now)					
TableChartPivot 6 rows2.6s					
#	AGE_GROUP	# TOTAL_APPOINTMENTS	# NO_SHOW_COUNT	# NO_SHOW_RATE_PCT	
1	Teen	6343	1690	26.64	
2	Young Adult	16731	4124	24.65	
3	Adult	29381	6180	21.03	
4	Child	21037	4307	20.47	
5	Middle Age	22635	3786	16.73	
6	Senior	14402	2233	15.50	

Results (just now)				
Table	Chart	Pivot	2 rows	107ms
	SMS_LABEL	# TOTAL_APPOINTMENTS	# NO_SHOW_COUNT	# NO_SHOW_RATE_PCT
1	No SMS	75045	12536	16.70
2	SMS Sent	35484	9784	27.57

Results (just now)				
Table	Chart	Pivot	81 rows	100ms
	NEIGHBOURHOOD	# TOTAL_APPOINTMENTS	# NO_SHOW_COUNT	# NO_SHOW_RATE_PCT
1	JARDIM CAMBURI	7717	1465	18.98
2	MARIA ORTIZ	5806	1220	21.01
3	ITARARÉ	3514	923	26.27
4	RESISTÊNCIA	4432	906	20.44
5	CENTRO	3334	703	21.09
6	JESUS DE NAZARETH	2853	696	24.40
7	JARDIM DA PENHA	3878	631	16.27
8	CARATOÍRA	2565	591	23.04
9	TABUAZEIRO	3132	573	18.30
10	BONFIM	2773	550	19.83
11	ILHA DO PRÍNCIPE	2266	532	23.48

Task 13: Connect Snowflake with Power BI

Connect Power BI Desktop to Snowflake and create a dashboard.

Dashboard title: Healthcare Appointment No-Show Dashboard

Create these DAX measures.

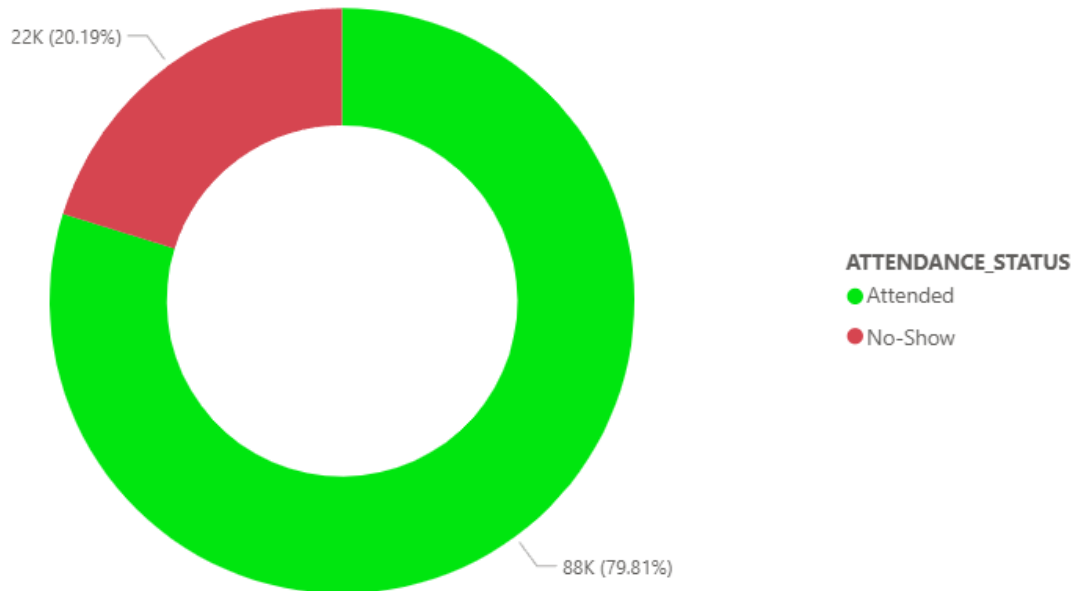
- Total Appointments
- Missed Appointments
- Visited Appointments
- No Show Rate
- Average Waiting Days
- Attendance Status Distribution -> Donut Chart
- No-Show by Age Group -> Clustered Column Chart

Healthcare Appointment No-Show Dashboard

ANS:



Attendance Status Distribution



No-Show by Age Group

